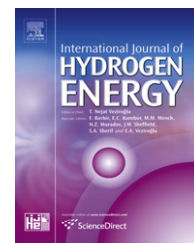


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Experimental and numerical study of a magnesium hydride tank

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ARTICLE INFO

Article history:

Received 10 July 2009

Received in revised form

4 March 2010

Accepted 11 March 2010

Available online 24 April 2010

Keywords:

Hydrogen storage

Magnesium hydride

Heat transfer

Numerical modeling

ABSTRACT

A small-scale experimental magnesium hydride tank was designed and tested to illustrate the feasibility of hydrogen storage in magnesium hydride. A prototype of the tank was filled with 123 g of previously ball-milled and doped MgH_2 . About 80 nl of hydrogen can be reversibly stored at a pressure less than 1 MPa. However, owing to the fact that the heat of a reaction limits the absorption and desorption processes, these latter are slowed down. To have a better understanding of the heat and mass transfers in the tank, a numerical model was developed using the Fluent software. The numerical simulations of hydrogen sorption are found to be in good agreement with the experimental results. During desorption, as an example, the reaction occurs locally and progresses from the tank walls towards the core. Low thermal conductivity of MgH_2 powder and slow heat evacuation (or supply) are identified to be responsible for the increase in the tank (un)loading duration.

A second configuration of the tank was tested with a heat exchanger and a new compacted material, composed of MgH_2 and expanded natural graphite (ENG), to increase thermal conductivity. Due to a better controlled heat transfer in the tank, this system can absorb 100 nl H_2 in only 25 min.

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1. Introduction

Hydrogen storage can be considered as a key factor in the development of a hydrogen economy. Among the technical solutions that are currently being considered, hydrogen storage in magnesium hydride (MgH_2) is a very promising technique for numerous reasons. Indeed, magnesium is abundant, relatively cheap, life-friendly and has a weight storage capacity of 7.6%, (one of the best among reversible metal hydrides).

The reaction between magnesium and hydrogen occurs in the temperature range of 250–370 °C and has very endo- or exo-thermic characteristics depending on whether desorption or absorption is considered. Therefore, heat should be

removed from the tank during absorption and supplied during desorption. In the past, this fact has discouraged several attempts to design a MgH_2 tank. However, thermodynamic stability can also be an advantage in terms of security: leakage in such a tank leads to a decrease in temperature, which results in the cessation of the hydrogen desorption reaction. Thanks to the progress that has been made in the MgH_2 synthesis [1–4], fast sorption kinetics can be obtained at reasonable temperatures.

Very few MgH_2 tanks have been studied. Buchner et al. [5] tested Mg_2Ni in a tank in the 80s, and Marty et al. [6] and Askri et al. [7] have studied it numerically.

This paper focuses on a combined experimental and numerical study of a magnesium hydride tank. The first

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doi:10.1016/j.ijhydene.2010.03.057

Nomenclature

C_p	specific heat capacity, J/kg/K
E_a	activation energy, J/mol
ΔH°	molar enthalpy of reaction at reference conditions, J/mol
ΔH	molar enthalpy of reaction under real conditions, J/mol
K	permeability ($\rightarrow \nabla P = \mu/K \cdot \vec{V}$), m^2
k	kinetic coefficient, s^{-1}
k_0	kinetic law coefficient, s^{-1}
L_{heat}	typical heat diffusion length, m
L_{gas}	typical gas diffusion length, m
μ	dynamic viscosity of hydrogen, Pa.s
M	molar mass of hydrogen, kg/mol
P	hydrogen pressure, Pa
P_{in}	inlet pressure of hydrogen, Pa
P_0	reference pressure = 101.325 kPa
Pe	Peclet number = $Pe = V \cdot L_{\text{gas}} \cdot \rho_g \cdot C_{pg} / \lambda$
R	universal gas constant = $[J \cdot K^{-1} \cdot \text{mol}^{-1}]$
R_{th}	thermal contact resistance, $K \cdot m^2/W$
S	heat source, W/m^3
ΔS°	molar entropy of reaction at standard conditions, $J/mol \cdot K$
T	temperature, K
V	superficial velocity, m/s

V_r	actual gas velocity between the powder particles ($V = \varepsilon V_r$), m/s
wt	maximum mass content of hydrogen in the metal, %

Greek letters

α	hydriding fraction ($0 < \alpha < 1$): $\alpha = n_{\text{MgH}_2} / (n_{\text{MgH}_2} + n_{\text{Mg}})$
α_{20}	hydriding fraction after 20 min of reaction at a given temperature
α'	relative hydriding fraction: $\alpha' = \alpha / \alpha_{20}$
λ	effective thermal conductivity, $W/m \cdot K$
ϕ	heat flux density, $W \cdot m^{-2}$
ρ	density, kg/m^3
ψ	pressure loss coefficient inside the filter, m^2
$d\alpha/dt$	hydriding velocity, s^{-1}
ε	porosity

subscripts

g	gas, hydrogen
m	bulk metal
heat	heat transfer
gas	gas transfer
eq	equilibrium
abs	absorption
des	desorption

section presents a small experimental tank, and its performances. The second section refers to the numerical modeling that has been specifically developed for this tank. The experimental results and numerical simulations are also compared in this section. The last section then describes a new configuration of the tank equipped with an improved thermal conduction porous bed and combined with a heat exchanger designed to increase the rate of hydrogen sorption.

2. Experimental study of a MgH_2 tank

2.1. Experimental set-up

A magnesium hydride tank for hydrogen storage cannot be designed as a simple low-pressure vessel. Because of the thermodynamic stability of MgH_2 , it has to be resistant to up to 400°C in temperature and up to 1.5 MPa in pressure. Moreover, the tank has to be designed to prevent the spreading of the hydride powder and to endure mechanical stresses resulting from a possible powder dilatation when hydrogenation occurs.

A small experimental tank has been designed to test the feasibility of hydrogen storage in MgH_2 . Its cylindrical inner volume is 270 cm^3 . A 316 L-grade stainless steel vessel contains the powder, and an aluminum shell surrounds the vessel to homogenize the temperature. A regulated heating coil ensures appropriate temperature conditions and provides the necessary heat to desorb hydrogen. During absorption, heat is evacuated by conduction through the tank walls, the tank housing and then by natural convection of air around the

latter. The tank is equipped with thermocouples placed in the powder bed, a pressure sensor and an inlet and outlet mass flow meter. The locations of the six thermocouples (TC1–TC6) are reported in Fig. 1.

The inner volume was filled with 123 g of ball-milled MgH_2 containing 4% (atomic) of Ti–V–Cr additives. This material

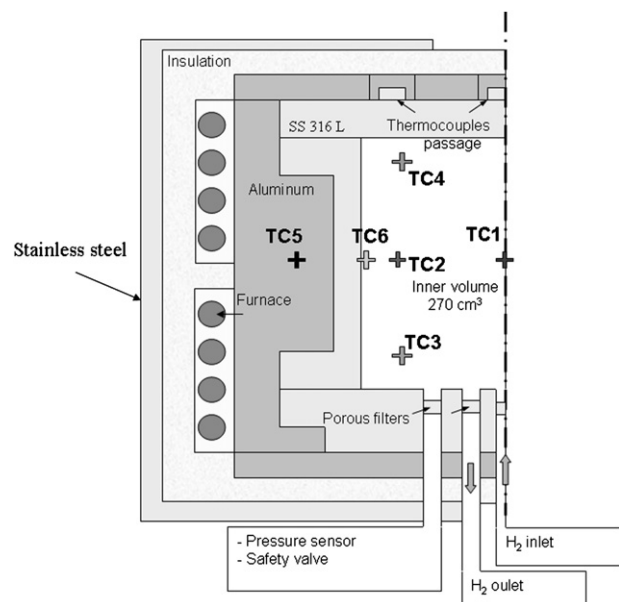


Fig. 1 – Sketch of the experimental magnesium hydride tank.

has been prepared by the McPhy Energy Company, using a ball-miller to deliver 1-kg batches of activated magnesium hydride [4].

2.2. Desorption test

As the material in the tank is hydrogenated during the synthesis process, a desorption procedure has to be performed first. To do so, the tank is initially heated up to 340 °C. At this temperature, the hydrogen pressure is higher than 0.45 MPa. At $t = 0$, once temperature is found to be homogeneously distributed, the exhaust valve is opened and hydrogen is released at atmospheric pressure (Fig. 2).

The porous filter prevents powder from spreading out of the tank. Consequently, the pressure in the powder bed is not equal to atmospheric pressure. During desorption, the hydrogen mass flow rate decreases. Then, the pressure loss through the filter decreases as well.

During the first few minutes of the reaction, the temperature inside the powder bed decreases and then stabilizes near the equilibrium temperature corresponding to the hydrogen pressure. After 1 h, two different behaviors are observed in the temperature evolution:

- the core temperature (TC1) decreases drastically until the end of the desorption and suddenly reaches a value close to the set-point temperature, and
- other local temperatures remain constant or increase slightly until the end of the desorption experiment.

These behaviors can be interpreted by the fact that the desorption reaction occurs locally in the powder bed and is therefore not homogeneous. At the very beginning of the reaction, the entire hydride bed desorbs hydrogen. Due to the endothermic characteristic of the desorption reaction, the temperature decreases to an equilibrium temperature. After having reached this temperature, no further reaction occurs except at the regions close to the walls, which benefit from the heat generated by the external heater.

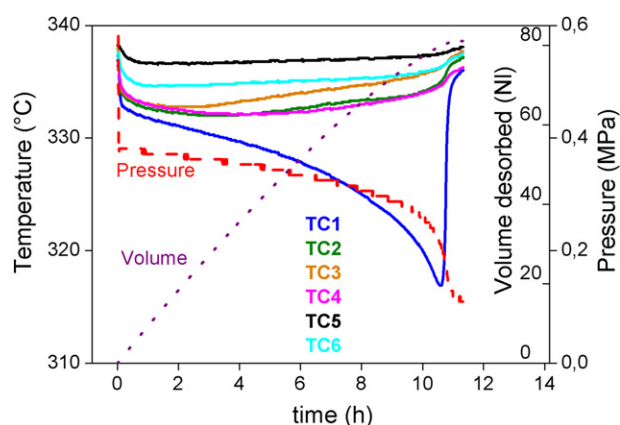


Fig. 2 – Experimental evolution of temperatures, pressure and hydrogen volume desorbed by the tank during desorption achieved at initial and set-point conditions of 340 °C and 0.1 MPa.

Conversely, the core region cannot receive the heat from the heater and is therefore confined to equilibrium. Owing to the pressure decrease during the process, the temperature in the center decreases until the last minutes of desorption. It is necessary to simply maintain the temperature of the external heater and to wait until the entire bed core has desorbed its hydrogen content. Once the entire powder bed has desorbed, the reaction stops and the powder bed reaches the set-point temperature. After 11 h, 80 nl of hydrogen is desorbed.

2.3. Absorption test

The tank is initially heated to 300 °C. The hydrogen pressure is set at 0.1 MPa, a value less than the equilibrium pressure. At $t = 0$, the inlet valve shown in Fig. 1 is opened.

Hydrogen is supplied at 0.77 MPa, but the porous filter generates a pressure loss so that the resulting pressure in the tank varies with the absorbed hydrogen mass flow rate during absorption.

During the first 5 min, temperature rises suddenly in the whole tank (Fig. 3). Then, the core temperature (TC1) rises during 30 min and stabilizes at the equilibrium temperature, corresponding to a pressure of 0.77 MPa, while temperatures recorded by TC2, TC3, TC4, and TC6 in other regions of the powder bed decrease progressively to reach the set temperature after 150 min. TC1 decreases suddenly after 140 min to reach the temperature set-point.

Absorption stops after 160 min, at which point almost 80 nl of H₂ has been absorbed.

The temperature evolutions can be interpreted as in the previous section: during the first 30 min, the powder located near the walls undergoes a reaction as the heat of the reaction is easily removed through the tank wall. During most of the experiment, the core cannot transfer heat, thus resulting in no reaction taking place in this specific region. The core temperature remains at the equilibrium temperature, corresponding to the pressure inside the tank. The center part is the last region of the hydride bed that absorbs hydrogen.

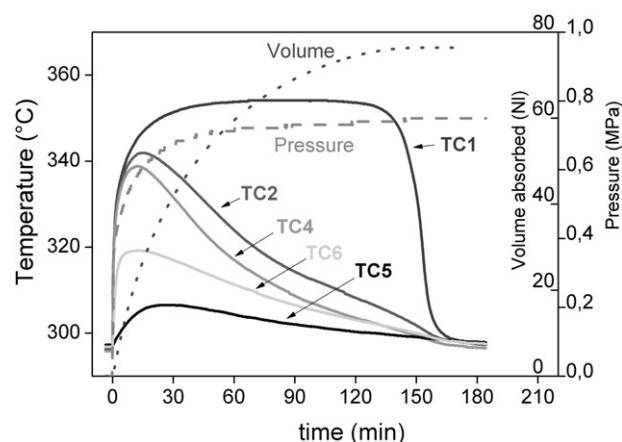


Fig. 3 – Experimental evolution of temperatures, pressure and hydrogen volume absorbed by the tank during an absorption achieved at initial and set conditions of 300 °C and 0.77 MPa.

Similarly to the desorption process, the absorption behavior can be explained by a progression of a reaction front from the tank walls towards the core of the powder bed. For both processes, the reaction time is much longer than it would have been for a small powder sample under the same temperature conditions. Absorption is faster than desorption because of the larger difference between equilibrium temperature and the wall temperature during absorption, leading to a more efficient heat transfer during an absorption test.

3. Numerical modeling

3.1. Parametric investigations

With the aim of confirming our interpretation, we designed a numerical model of a magnesium hydride tank to carry out our investigations. The first step of the model design involves the determination of physical and chemical parameters of the system, namely thermodynamic and kinetic parameters of activated MgH_2 using a volumetric Sievert apparatus (HERA C2-3000). The thermal physical characteristics of the powder were also determined.

3.1.1. Thermodynamic parameters

Van't Hoff's law links the equilibrium pressure to temperature:

$$\frac{P_{\text{eq}}}{P_0} = e^{(\Delta H^\circ / RT) - (\Delta S^\circ / R)} \quad (1)$$

Pressure Composition Temperature (PCT) diagrams have been recorded for temperatures ranging from 280 °C to 350 °C. As shown in Fig. 4a, the pressure equilibrium plateau is fairly constant for magnesium hydride at a given temperature. The differences between absorption and desorption values come from the stress induced by hydriding rate change in powder grains [8]. The maximum hydrogen capacity is reduced to about 6.5 wt.% due to the addition of the Ti–V–Cr alloy.

To determine the values of ΔH° and ΔS° , the quantity $\ln(P/P_{\text{eq}})$ was plotted as a function of $1/T$. Doing so yields values of $\Delta H^\circ = -75000 \text{ J/mol H}_2$ and $\Delta S^\circ = -135.6 \text{ J/K/mol H}_2$ for absorption and $\Delta H^\circ = -75500 \text{ J/mol H}_2$ and $\Delta S^\circ = -135.6 \text{ J/K/mol H}_2$ for desorption. These values are coherent with data found by Liang et al. [9].

As usual, we see in Fig. 4b that the curve showing the equilibrium pressure plateau vs. temperature separates Mg and MgH_2 domains as follows: a powder placed under conditions above the absorption curve will form MgH_2 , whereas if it is placed under conditions corresponding to the region below the curve, it will form Mg metal.

3.1.2. Kinetic law

Investigations of kinetic laws of ball-milled MgH_2 are complex since several mechanisms simultaneously occur during the hydrogenation process of a grain. Even temperature acts on the limiting mechanism that controls the kinetic law [10,11]. Since our goal is only to describe the kinetics, no interpretation of the obtained laws will be made.

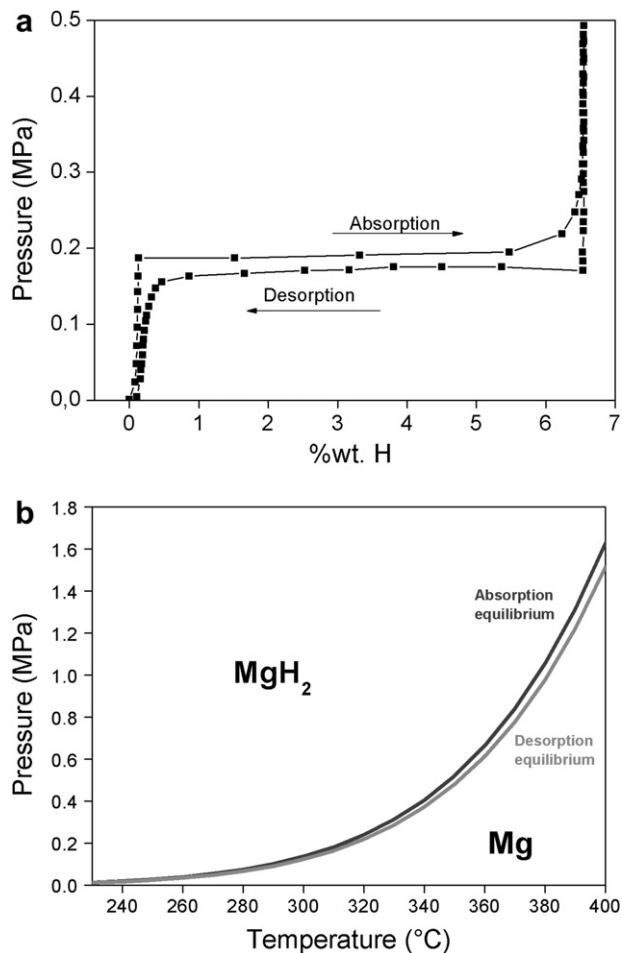


Fig. 4 – Equilibrium pressure of ball-milled and doped MgH_2 : a) isothermal pressure composition (PCT) at 300 °C; and b) equilibrium curves separating Mg and MgH_2 domains in a pressure–temperature diagram.

Absorption kinetics were measured at pressures of 1 MPa and 0.2 MPa in the temperature range of 200–350 °C. Under these experimental conditions, we observed that most of the magnesium was hydrogenated within the first 20 min of the experiment. SEM observations of the powder showed that a small fraction of the tested sample had been insufficiently ball-milled; this explains why more than 20 min were required to hydrogenate the powder since it contains a few larger particles. It is therefore convenient to introduce the variable $\alpha' = \alpha/\alpha_{20}$ where α is the hydrogenated fraction and α_{20} is its value at $t = 20 \text{ min}$. Kinetic measurements were linearized using different fitting laws corresponding to mechanisms that could limit the absorption process [12]. Thus, for the two pressures that were investigated, an Avrami-type equation with an exponent of $\theta = 2$ best fits the time evolution of the hydriding fraction α' .

$$(-\ln(1 - \alpha'))^\theta = k(T, P) \cdot t \quad (2)$$

Using the exponent $\theta = 2$, the kinetic plots shown in Fig. 5a for $P = 1 \text{ MPa}$ are linear for more than 90% of the reaction range.

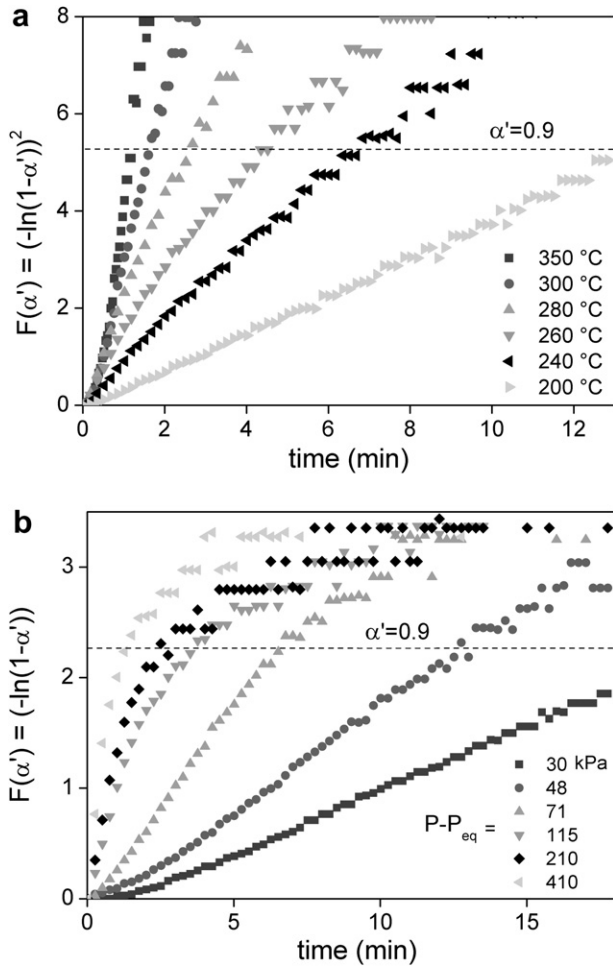


Fig. 5 – Linearization of absorption kinetics using an Avrami-type law for conditions far from (a) and close to (b) equilibrium.

Fig. 5b shows kinetics measured at 300 °C in the pressure range of 30 kPa–410 kPa above equilibrium pressure. We see that for conditions of pressure and temperature close to equilibrium ($P < 2 P_{eq}$), the Avrami equation requires an exponent of $\theta = 1$ to satisfy the linear dependence of Eq. (2) with time.

Desorption kinetics were measured at 15 kPa in the temperature range of 270–350 °C. Fig. 6 shows the plot of $(-\ln(\alpha'))^{1/2}$ versus time. Kinetic plots were found to be linear over at least 90% of the reaction range. An exponent of $\theta = 1/2$ in the Avrami equation best describes the time evolution of the hydrogenation fraction α' .

The effect of pressure on kinetics was also investigated. For an absorption test achieved at 300 °C, Fig. 7a shows that the evolution of the coefficient k is linear to the relative pressure difference ($P - P_{eq}$).

For desorption at 360 °C, Fig. 7b shows that the evolution of the coefficient k is linear to $\ln(P_{eq}/P)$.

The impact of temperature on kinetics was determined using the Arrhenius law:

$$k(T, P) = k_0 \cdot f(P) \cdot e^{-E_a/RT} \quad (3)$$

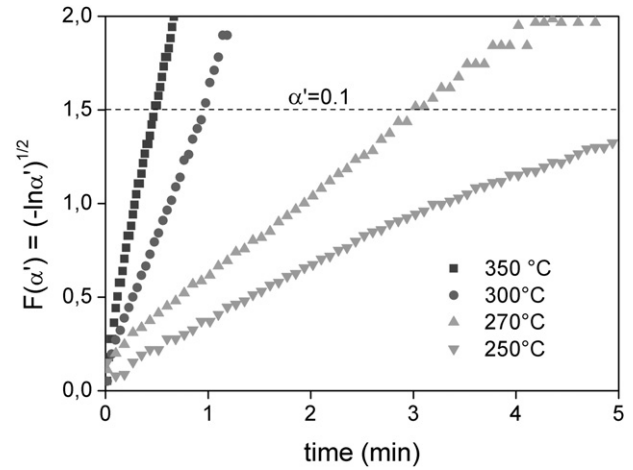


Fig. 6 – Linearization of desorption kinetics using an Avrami-type law at 15 kPa.

with $f(P) = (P/P_{eq}) - 1$ for absorption and $f(P) = \ln(P/P_{eq})$ for desorption.

Plotting $\ln(k) - \ln((P/P_{eq}) - 1)$ vs. $1/T$ for kinetics analysis, measured at 1 MPa and 0.2 MPa, allows us to determine E_a and k_0 during absorption.

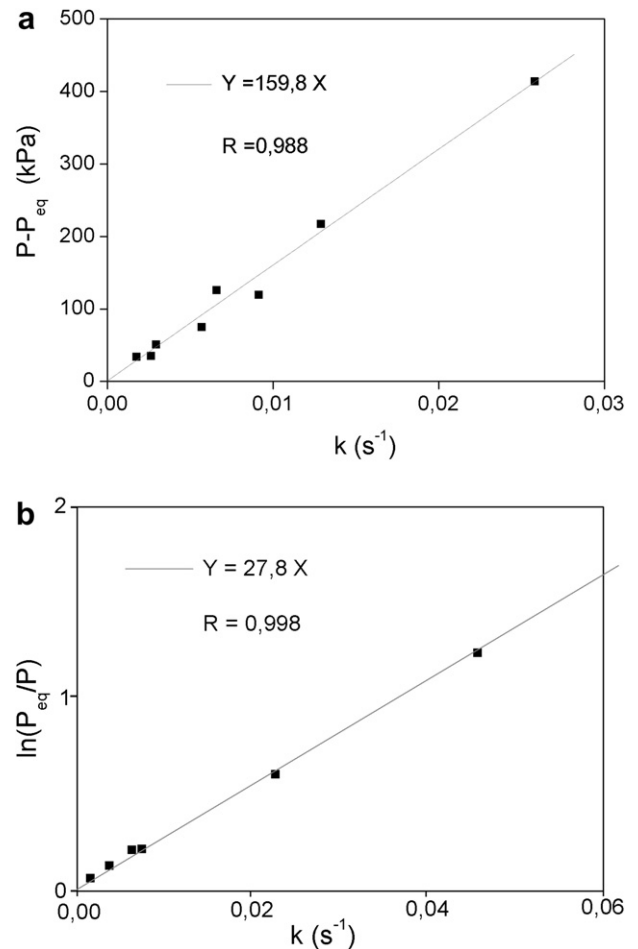


Fig. 7 – Kinetic coefficient vs. hydrogen pressure for absorption at 300 °C (a) and desorption at 360 °C (b).

Table 1 – Activation energy E_a and coefficient k_0 measured for absorption (1 and 0.2 MPa) and desorption (15 kPa).

	Pressure (MPa)	E_a (kJ/K/mol H_2)	k_0 (s^{-1})
Absorption	1	132	9.8×10^9
	0.2	128	1.14×10^{10}
Desorption	0.015	41	10

The same plot is made with $\ln(k) - \ln(\ln(P_{eq}/P))$ for desorption kinetics measured at 15 kPa.

Table 1 summarizes the values obtained for the activation energy E_a and the coefficient k_0 .

The energy of activation, E_a , and the coefficient k_0 are quite independent of pressure. For absorption, the following mean values will be retained for the numerical applications: $E_a = 130$ kJ/K/mol H_2 , $k_0 = 10^{10} s^{-1}$.

Using the Arrhenius law, an implicit kinetics law can be deduced from the previous results. Table 2 presents these results.

3.1.3. Thermal conductivity

Due to the exo- or endo-thermic nature of the reaction between hydrogen and magnesium, it is important to determine the thermal conductivity of the powder. In fact, the efficiency of the heat transfer will have an important impact on the temperature of the powder and, subsequently, on the kinetics of the absorption and desorption processes.

Thermal conductivity of Mg–MgH₂ systems has been studied by Kapischke et al. [13], but this parameter is extrinsic and strongly depends on the preparation conditions of the material. Moreover, porosity, annealing condition, and microstructure of grains may affect the thermal conductivity. For this reason, the thermal conductivity of the powder has to be measured in situ, i.e., inside the tank.

To perform this measurement, the tank is initially set at room temperature. Hydrogen pressure is kept constant during this experiment. The heater is then set to 300 °C and reaches this set-point temperature after a few minutes. Owing to thermal inertia, the aluminum shell temperature increases slowly. The delay between the heater and shell temperatures is a function of the thermal contact resistance R_{th} between

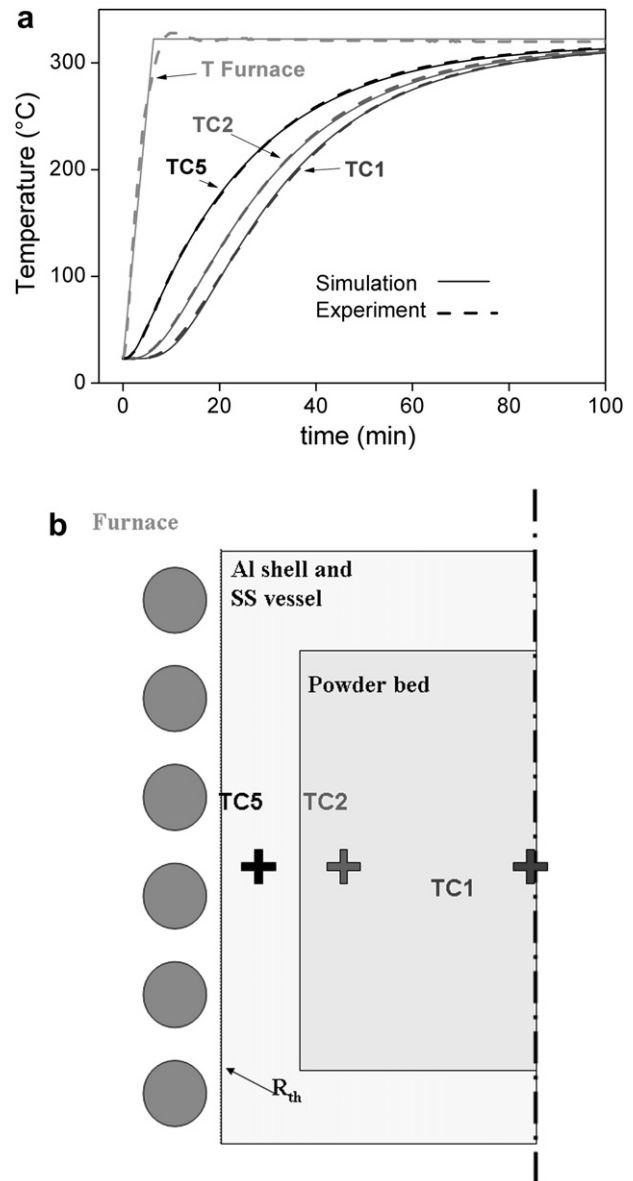


Fig. 8 – Experimental and simulated rising temperature of the tank to fit the powder bed thermal conductivity (a); thermocouple locations inside the tank (b).

Table 2 – Kinetics laws for hydrogen sorption.

	Absorption	Desorption
$P > 2 P_{eq}$	$\frac{d\alpha'}{dt} = k_0 \cdot \frac{\alpha' - 1}{2 \cdot \ln(1 - \alpha')} \cdot \left(\frac{P}{P_{eq}} - 1 \right) \cdot e^{-E_a/Rt} \quad (12)$	
$P_{eq} < P < 2P_{eq}$	$\frac{d\alpha'}{dt} = k_0 \cdot (1 - \alpha') \cdot \left(\frac{P}{P_{eq}} - 1 \right) \cdot e^{-E_a/Rt} \quad (13)$	
$P < P_{eq}$		$\frac{d\alpha'}{dt} = 2k_0 \frac{\alpha'}{(-\ln \alpha')^{1/2}} \ln \left(\frac{P_{eq}}{P} \right) \cdot e^{-E_a/Rt} \quad (14)$

those two elements (Fig. 8a). Another temperature delay can be measured between two thermocouples in the powder bed, one placed at the center (TC1) and the other at the radius $r = 29$ mm (TC2) (Fig. 8b).

A simple 2D axisymmetric thermal conduction simulation allows us to estimate the thermal conductivity of the powder by trial and error.

Numerical predictions fit well with the experimental plots for $1/R_{th} = 47$ W/m²/K and $\lambda = 0.48$ W/m/K.

3.2. Equations

The finite volume technique implemented in the commercial code Fluent was used to model the tank. A two-dimensional axisymmetric model was used.

The main assumptions are similar to those used in previous numerical studies done on LaNi₅ [14–18]:

- hydrogen behaves like an ideal gas;
- local thermal equilibrium between gas and powder particles is assumed; and
- radiative heat transfer is neglected, although it may affect the temperature field of a powder bed above 300 °C, as mentioned by Askari et al. [7]. In fact, interaction between radiation and particles is complex as the radiation wavelength is quite comparable to the grain size.

3.2.1. Heat transfer

Assuming thermal equilibrium is reached between hydrogen and porous MgH₂, the energy equation becomes

$$\varepsilon \cdot \rho_g \cdot C_{pg} \cdot \frac{\partial T}{\partial t} + (1 - \varepsilon) \cdot \rho_m \cdot C_{pm} \cdot \frac{\partial T}{\partial t} + \varepsilon \cdot \rho_g \cdot C_{pg} \cdot \nabla(T \cdot \vec{V}_r) = \lambda \cdot \nabla^2 T + S \quad (4)$$

where λ is the effective thermal conductivity of the porous bed. Without any experimental value of λ available, it is tempting to use the classical mixing law:

$$\lambda = \varepsilon \cdot \lambda_g + (1 - \varepsilon) \cdot \lambda_m \quad (5)$$

where λ_g is the thermal conductivity of hydrogen and λ_m is that of the mixture of magnesium metal, magnesium hydride and additives. However, experimental results deduced from the previous section showed that the value given by Eq. (5) is eight times smaller than the actual value. We have chosen to use the experimental value of λ for the rest of the work presented in the following sections.

Introduction of the averaged quantities:

$$\rho \cdot C_p = \varepsilon \cdot \rho_g \cdot C_{pg} + (1 - \varepsilon) \cdot \rho_m \cdot C_{pm} \quad (6a)$$

$$\vec{V} = \varepsilon \cdot \vec{V}_r \quad (6b)$$

allows a simplification of the energy equation, which becomes

$$\rho \cdot C_p \cdot \frac{\partial T}{\partial t} + \rho_g \cdot C_{pg} \cdot \nabla(T \cdot \vec{V}) = \lambda \cdot \nabla^2 T + \frac{\rho_m \cdot wt \cdot (1 - \varepsilon)}{M} \cdot \frac{d\alpha}{dt} \cdot \Delta H^\circ \quad (7)$$

The order of magnitude of α_{20} is very close to unity for temperatures larger than 300 °C. From now throughout the

rest of the manuscript, α' will be assimilated to α . Concerning the reaction between magnesium metal and hydrogen, the enthalpy of the reaction under real conditions (ΔH) is assumed to be equal to the enthalpy under referenced conditions (ΔH°). Therefore, the source term at the right-hand side of Eq. (7), which is proportional to the reaction rate, becomes

$$S = \frac{\rho_m \cdot wt \cdot (1 - \varepsilon)}{M} \cdot \frac{d\alpha}{dt} \cdot \Delta H^\circ \quad (8)$$

Thermodynamic and kinetic aspects are treated by calculating P_{eq} , $d\alpha/dt$ and α in each cell. Doped magnesium hydride kinetics laws are given by Eqs. (4)–(6). The equilibrium pressure is computed using Eq. (1).

3.2.2. Mass transfer

The possibility to avoid calculating the hydrogen flow during sorption in magnesium hydride tanks has been studied in a previous publication [19]. From this work, three conditions have to be satisfied to neglect the influence of the gas flow:

- convective heat transfer has to be negligible compared to heat diffusion, which implies that the Peclet number $Pe = (V \cdot L_{gas} \cdot \rho_g \cdot C_{pg})/\lambda$ should be much smaller than unity;
- sensible heat of hydrogen penetration in the tank has to be weak compared to the enthalpy of the reaction; and
- pressure variations induced by the flow have no impact on hydriding velocity if the criteria $C \gg 1$ and $N \ll 1$ are met, C and N being defined as follows:

$$C = \frac{\partial P_{eq}}{\partial T} \cdot \frac{L_{heat}^2 \cdot (1 - \varepsilon) \cdot wt \cdot \rho_m \cdot \Delta H^\circ}{\lambda \cdot M} \cdot k_0 \cdot \frac{e^{-Ea/RT}}{P_{eq}} \quad (9)$$

and

$$N = \frac{\lambda \cdot M \cdot L_{gas}^2 \cdot \mu}{\frac{\partial P_{eq}}{\partial T} \cdot \Delta H^\circ \cdot \rho_g \cdot K \cdot L_{heat}^2} \quad (10)$$

where L_{gas} and L_{heat} are the typical dimensions over which pressure and temperature vary, respectively. In our case, L_{gas} is equal to the axial height of the tank (0.07 m) and L_{heat} is the tank radius (0.035 m). An order of magnitude of the velocity can be deduced from the value of the mean mass flow rate, thereby resulting in $V = 3 \times 10^{-4}$ m s⁻¹. This allows us to calculate the Peclet number, yielding a value of $Pe = 0.2$. Although this value is only slightly less than unity, heat convection will be neglected in comparison to heat conduction.

One can also wonder about the cooling effect of the cold hydrogen that is entering the tank; in our facility, hydrogen entering the tank is heated over a few centimeters by pipes connected to the stainless steel shell. Therefore, hydrogen does not provide a significant cooling of the powder bed.

Table 3 presents the values of the physical parameters that are necessary for the computation. Computing C and N gives $C > 1000$, which means heat diffusion is limits kinetics and thus controls the hydriding process. Furthermore, N is found to be smaller than 10^{-3} , which means that pressure variations do not affect the hydrogenation process.

Table 3 – Operating conditions and physical parameters of magnesium hydride and hydrogen at 600 K.

Doped magnesium hydride:	Operating parameters:	Hydrogen properties:
$k_0 = 10^{10} \text{ s}^{-1}$ $E_a = 130 \text{ kJ}$ $\Delta H^\circ = -75000 \text{ J/mol}$ $\Delta S^\circ = -135.6 \text{ J/K/mol}$ $\rho_m = 1800 \text{ kg/m}^3$ $\lambda = 0.48 \text{ W/m/K}$ $\varepsilon = 0.74$ $\text{wt} = 0.06 \text{ (6 wt.\% H)}$ $K = 5,75 \times 10^{-14} \text{ (m}^2\text{) (Ergun equation)}$ $C_p = 1545 \text{ J/kg/K}$	$T = 630 \text{ K}$ $P_{\text{in}} = P_{\text{eq}} = 0.8 \text{ MPa}$ $L_{\text{gas}} = 0.07 \text{ m}$ $L_{\text{heat}} = 0.035 \text{ m}$	$\mu = 8.9 \times 10^{-6} \text{ Pa s}$ $M = 0.002 \text{ kg/mol}$ $C_{pg} = 14000 \text{ J/kg/K}$ $\lambda_g = 0.24 \text{ W/m/K}$

From the above results, hydrogen flow can be neglected in our numerical modeling. Pressure is then considered to be homogeneous in space but to vary with time. The pressure loss generated by the porous filter is considered to be linear with the mass flow rate of hydrogen: $\Delta P = \psi \cdot m_{\text{inlet}}$ with $\psi = 9 \times 10^9 \text{ Pa s/kg}$. Pressure inside the powder bed is computed via a subroutine based on a mass balance between the hydrogen entering the tank through the filter, hydrogen consumed by the reaction, and hydrogen present in the porosities.

3.3. Mesh and boundary conditions

Three parts of the tank were meshed: the powder bed, the stainless steel vessel, and the aluminum shell. Approximately 10 000 triangular meshes were used for the vessel as well as for the shell, and 12 000 rectangular meshes were used for the powder bed.

In the experiment, the heater is supposed to be isothermal. As a consequence of the thermal contact resistance observed between aluminum and the heater, the heat flux density φ through the external shell wall depends on the heating temperature T_f and on the external wall temperature T_l of the shell as follows:

$$\varphi = \frac{T_f - T_l}{R_{th}} \quad (11)$$

4. Results

Numerical and experimental results for hydrogen absorption are compared in Fig. 9.

Computed evolutions of temperature, pressure and absorbed volume of hydrogen align with experimental plots relatively well. The loading time is correctly estimated, and the assumptions and the numerical model offer a good description of the hydrogenation process.

Simulation allows $d\alpha/dt$ and α to be known locally at any moment of the process. Fig. 10a depicts the reaction velocity $d\alpha/dt$ while Fig. 10b shows the hydriding fraction α in the MgH_2 powder bed at $t = 2150 \text{ s}$.

A hydrogenation front was observed as separating fully hydrogenated regions near the walls from regions in the core that had not yet been hydrogenated.

The time evolution of the reaction rate and temperature profiles are shown in Fig. 11 along the radius at mid-height. The reaction front moves from the periphery of the tank to its center (Fig. 11a). At a given time, the front shown in Fig. 11b separates a non-hydrogenated region, where temperature is constant and equal to the equilibrium temperature (near the center), from a hydrogenated region,

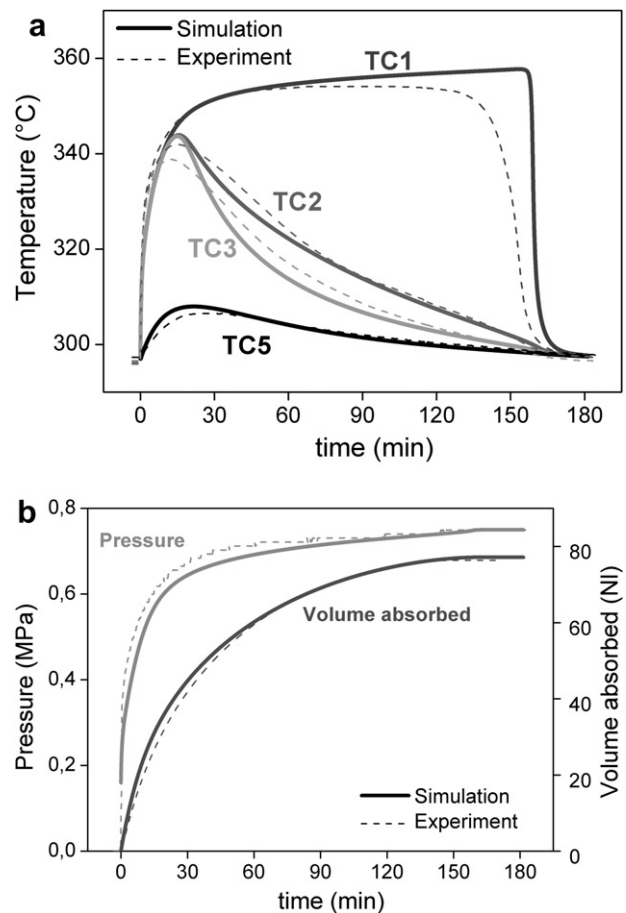


Fig. 9 – Simulated and experimental data for an absorption where the initial and the set-point conditions are 300 °C and 0.77 MPa: (a) temperature and (b) pressure and absorbed volume.

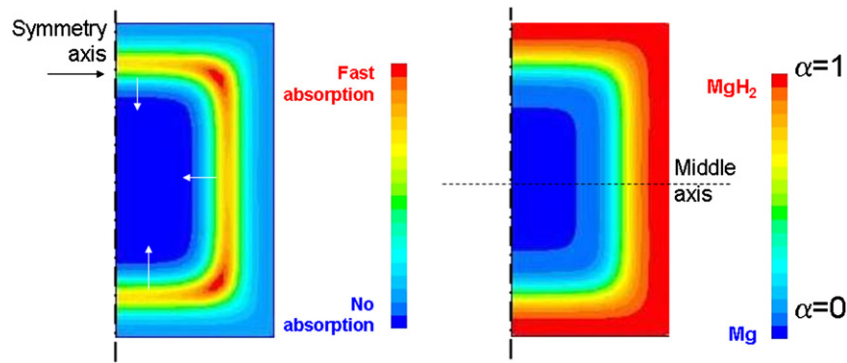


Fig. 10 – Computed maps of sorption velocity (a) and hydrogenation fraction (b) in the tank after 2150 s of absorption at a set-point of $P = 0.77$ MPa.

where temperature decreases to the wall temperature (≈ 300 °C).

The same behavior is observed when desorption is simulated.

The numerical simulation confirms that absorption and desorption processes in the tank occur locally, near

a narrow reaction front. The energy produced or consumed in the front is equal to the energy that is transferred through the tank. In fact, heat transfer is not sufficient to supply or remove the heat of the reaction from the whole powder bed within a few minutes: heat diffusion is the limiting factor for fast hydrogen storage in a magnesium hydride tank. The next paragraph describes an efficient solution to this problem.

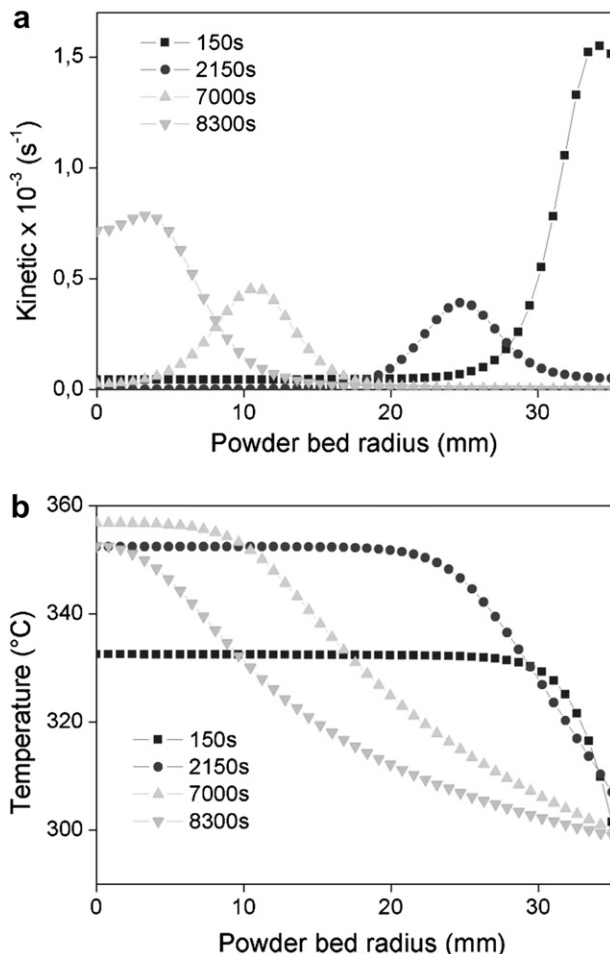


Fig. 11 – Computed profiles of absorption velocity (a) and temperature (b) along a radius at mid-height of the tank.

5. A new tank configuration for faster hydrogen sorption

To improve heat diffusion in the tank, a new material has been developed: MgH_2 has been mixed with 5 wt.% of expanded natural graphite (ENG). The resulting mixture was pressed in a cylindrical mold at 1 t/cm^2 to obtain a solid and compacted disc-shaped powder element [20].

Thermal conductivity was measured in both radial and axial directions in the CNRS-PROMES laboratory in Perpignan, France. The results are: $\lambda_{\text{radial}} = 4.2 \text{ W/m/K}$, $\lambda_{\text{axial}} = 1 \text{ W/m/K}$. Radial conductivity is ten times higher than that of the initial magnesium powder. Porosity is reduced by a factor of two with compression, resulting in an increase in the powder bed volumetric storage capacity.

A heat exchanger, adapted to the discs, was designed for the tank studied previously. It is composed of a central tube in which compressed air flows. Copper fins are alternatively stacked with discs. A hole at the center of the stack is filled with a cooling tube (Fig. 12).

Due to the reduced porosity of the material ($\epsilon = 0.31$), the tank now contains 156 g of MgH_2 . However, the low porosity induces a permeability that is too low ($K = 5.9 \times 10^{-16} \text{ m}^2$). The disc thickness has to be made lower than 20 mm to ensure that the same hydrogen pressure is exerted inside and outside of the discs.

In the present arrangement, disc thickness corresponds to the thickness of the reaction front in the previous configuration, i.e., 10 mm.

To improve the gas circulation inside the tank, a 1-mm free space was left for the radial circulation of hydrogen between each disc and the fin located just above (Fig. 12a). For the same

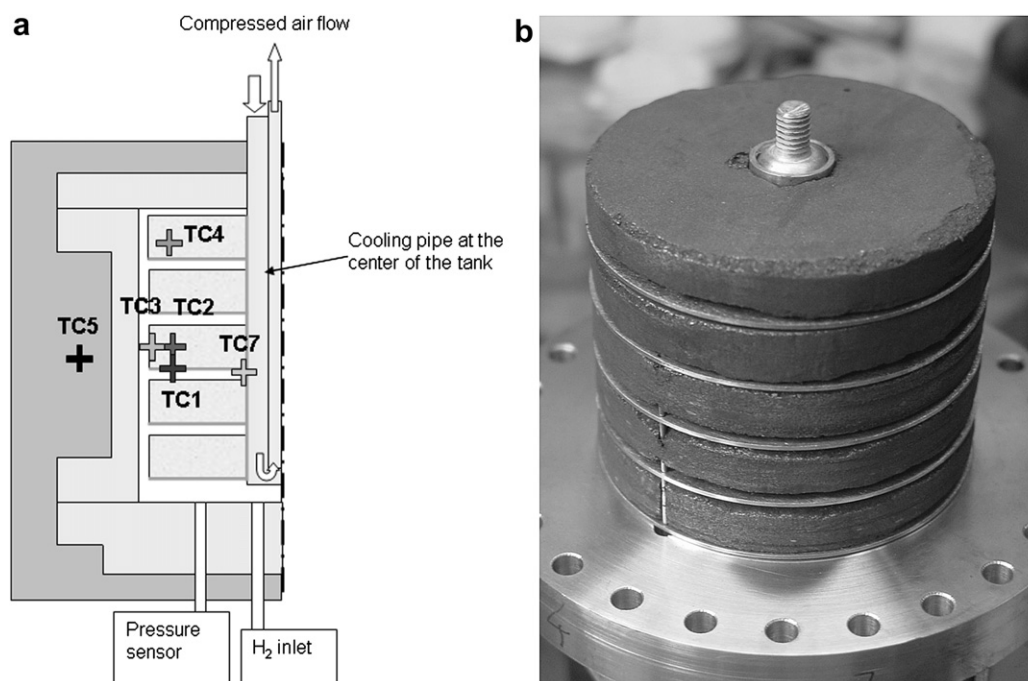


Fig. 12 – Sketch of the tank equipped with a copper fin heat exchanger fitting the compacted discs (a) and stack of $\text{MgH}_2 + 5 \text{ wt.\% ENG}$ (b).

reason, the diameter of both discs and fins is 1 mm less than the inner tank diameter.

Temperatures are measured at five points (TC1, TC2, TC3, TC4 and TC5), and hydrogen pressure is recorded around the discs. Mass flow rate is also recorded and integrated so that the volume of absorbed hydrogen can be determined.

Hydrogen absorption tests were performed at 300 °C and 1.1 MPa with a cooling air flow rate ranging from 0 to 22 l/min through the heat exchanger. Fig. 13 shows that almost 100 nl

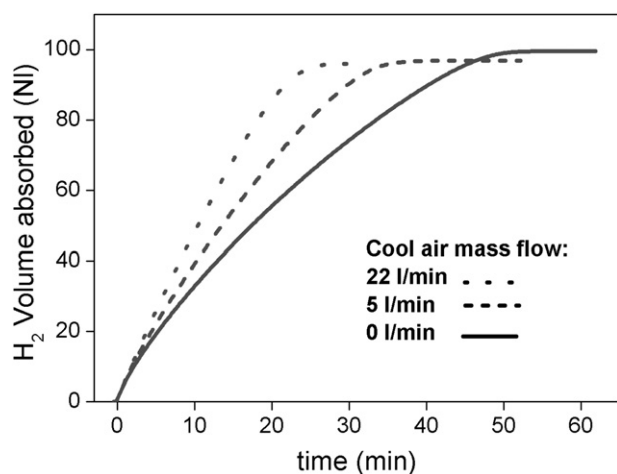


Fig. 13 – Absorbed hydrogen volume evolution during absorption achieved at 300 °C and 1.1 MPa with different air flow rates through the cooling heat exchanger.

of hydrogen is absorbed after 50 min without cooling. The introduction of a heat exchanger and compacted discs reduces the loading time by a factor of three compared to that obtained by the use of pure magnesium metal powder.

The impact of the air flow rate on the loading time is clear: increasing the air flow rate lowers the loading time. Only 25 min are necessary for the absorption of 95 nl of hydrogen with an air flow rate of 22 l/min.

Heat removal can be considered as the limiting phenomenon of the sorption process of an MgH_2 tank.

To validate the numerical model previously developed, desorption at 330 °C and 0.1 MPa, and absorption at 300 °C and 1.1 MPa with an air flow rate of 5 l/min were simulated with the data shown in Table 3. Computation of the parameters C (Eq. (9)) and N (Eq. (10)) shows that, in our case, the hydrogen flow can be neglected.

The experimental and simulation results are found to be in fair agreement with this model (Fig. 14a–d), thereby confirming the relevance of the assumptions used in our model.

This new tank configuration, with a heat exchanger and compacted magnesium hydride discs containing ENG, succeeds in reducing the loading time of both absorption and desorption processes. Heat transfer is markedly increased by copper fins and ENG. During absorption, heat removal occurs simultaneously through the vessel walls and the air flow. During desorption, the supplied heat has to diffuse entirely through the vessel walls, justifying why desorption is slower than absorption. The future prototypes of MgH_2 tanks will incorporate heaters directly placed into the powder bed to improve heat diffusion during desorption.

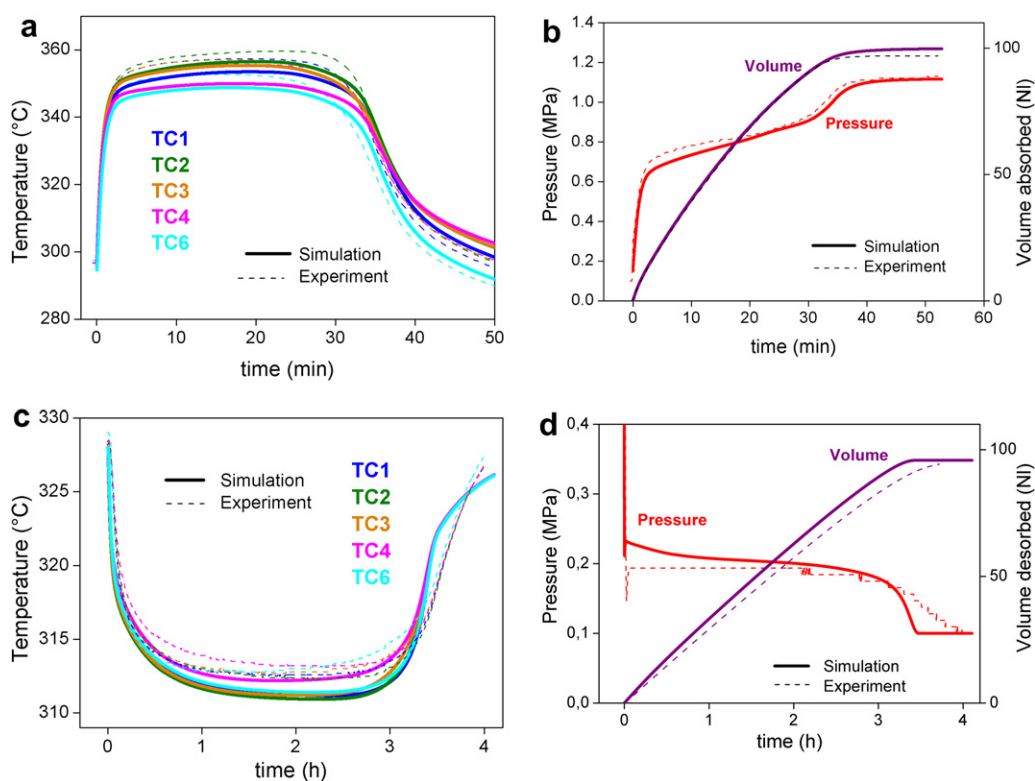


Fig. 14 – Comparison between experimental and numerical results obtained for the new tank geometry. (a–b) Absorption with 5 l/min of cool air at initial and set-point conditions of 300 °C and 1.1 MPa and (c–d) desorption at initial and set-point conditions of 330 °C and 0.1 MPa.

6. Conclusion

A small-scale experimental magnesium hydride tank has been designed and tested to prove the feasibility of hydrogen storage in magnesium hydride. A first configuration of the tank was filled with 123 g of ball-milled and doped MgH_2 . About 80 nl of hydrogen can be reversibly stored at a pressure lower than 1 MPa. The effective storage capacity of the material is 5.8 wt.% H_2 .

However, absorption and desorption are slow (2 h 30 min and 11 h, respectively) as a consequence of the limitation of heat diffusion. A numerical model, based on Fluent software, has been developed to understand the thermal and chemical behavior of the tank. Thanks to parametric measurements and to simplifications presented in a previous paper [19], simulations of hydrogen sorption in the tank were found to be in good agreement with experimental results. The behavior of the reacting powder bed was confirmed to be controlled by temperature. During absorption, the reaction occurs locally in a front moving from the tank walls to the inner core. A similar feature is observed during desorption.

Low thermal conductivity of MgH_2 powder and slow heat evacuation (or supply) have been identified to be responsible for the increase in the tank (un) loading time.

A second configuration of the tank was tested with a heat exchanger and a new compacted material composed of MgH_2 and expanded natural graphite (ENG) to increase the thermal

conductivity of the bed. An air flow through the central tube allowed the compacted powder bed to be cooled down efficiently. This system can absorb 100 nl of H_2 in only 25 min. For this configuration, the numerical model was found to be in perfect agreement with the experimental results.

Hydrogen storage in magnesium hydride appears to be very efficient and applicable. The production process on ball-milled magnesium hydride is mature and has been developed for use at the industrial scale by the McPhy Energy Company. In the future, the development of larger tanks will require an optimization of the heat transfer management and eventually heat storage to accumulate and recover the valuable heat that is released during absorption of magnesium hydrides.

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